PSYCH-UA 11: Statistics and Data Analysis for Research in Psychology

Spring 2026

Course Information

Time / Location:
Monday / Wednesday 12:30 - 1:45
31 Washington Pl (Silver Ctr), Room 411

Prerequisites for this course: None.

This class is worth 4 credits.

As of Fall 2023, students have now a variety of options to fulfill their statistics requirement:

- Statistics for Behav. Sci. (PSYCH-UA 10) and this course (PSYCH-UA 11), in any order.
- This course (PSYCH-UA 11) and Data Literacy (PSYCH-UA 8), in any order.
- This course (PSYCH-UA 11) and a quantitative advanced elective, in any order.
- Statistics for Behav. Sci. (PSYCH-UA 10) and Data Literacy (PSYCH-UA 8), in any order.
- Statistics for Behav. Sci. (PSYCH-UA 10) and a quantitative advanced elective, in any order.

Note that students can now fulfill their requirements without PSYCH-UA 11 (e.g., see the last 2 options).

Course Description

This course provides a deeper understanding of research methodology and statistical tests. Students perform their own data analysis and data visualization utilizing the software package R. Strongly recommended for students interested in psychological research.

Learning objectives

- Develop a functional understanding of statistics and its applications in psychological research
- Gain proficiency in using R for data analysis and visualization
- Understand and apply both parametric and nonparametric statistical methods
- Learn to critically evaluate and interpret statistical results in research contexts
- Develop skills in data manipulation, exploratory data analysis, and advanced statistical techniques

Instructor Information

Dr. Kelsey Moty (Pronouns: she/her)
Email: moty@nyu.edu
Office: Meyer Hall (6 Washington Pl.), Room 401

Office hours:
In-person: Mondays 2 - 3 pm (drop-in, no appointment necessary!)
Via Zoom: Appointments can be scheduled at: https://calendly.com/kelseymoty/office-hours

If none of these hours listed work for you, please reach out to me via email to schedule an appointment.

Class Assistant:
Avery Liang
Email: ll5234@nyu.edu
Office Hours: TBD

Course Materials

Textbook

While our lectures will not follow a particular textbook, I recommend the following online resources:

- R for Data Science by Hadley Wickham (https://r4ds.hadley.nz/) – abbreviated as RDS
- Learning Statistics with R by Danielle Navarro (https://learningstatisticswithr.com/book/) – abbreviated as LSR
- Experimentology by Michael C. Frank et al. (https://experimentology.io/) – abbreviated as E

Additional Readings

Any additional readings will be made accessible via Brightspace.

Other Resources

- **Databases, journal articles, and more**: Bobst Library (library.nyu.edu)
- **Assistance with strengthening your writing**: NYU Writing Center (nyu.mywconline.com)
- **Obtain 24/7 technology assistance**: IT Help Desk (nyu.edu/it/servicedesk)
- **Tutoring and academic skills workshops are offered by the University Learning Center**: www.nyu.edu/cas/clc
- **Support for personal and health-related issues at the Wellness Exchange**: (212) 443-9999 (24/7 confidential calls): www.nyu.edu/999

Course Policies

CAS Academic Policies can be found at https://cas.nyu.edu/academic-programs/bulletin/policies/academic-policies.html.

1. First and foremost, please, please, please ask questions.

Lectures are more interesting (for both you and me!) if I am not talking the entire time. You might think the answer is obvious to others, but there's a good chance that at least several other students in the class have the same question. I also welcome your opinions and observations and look forward to your contributions to lectures and discussions.

2. Update policy

The syllabus may change as necessary during the semester. Reasonable efforts (announcement in class and through email via Brightspace Announcements) will be made to update students if changes arise. Students are responsible for the current syllabus (if there are no changes) and any updates that might occur.

3. Attendance

Lecture
Attendance at lectures is **highly recommended, but not mandatory**. This class is designed for in-person learning, meaning your learning experience will likely be better if you attend class in-person. For example, I cannot answer questions asked via Zoom (I don't have enough screen space to keep the chat box open on my computer!)--but I can answer questions asked in-person.

That being said, all lectures will be available live via Zoom and will be recorded. Zoom links and recordings can be accessed via Brightspace.

If you do not attend the lecture live, you can catch up on the missed material by watching the recording.

Recitation
Recitation provides an additional opportunity to discuss and engage with the material in class. Attendance at recitation is **mandatory**. However, I understand that life happens, so you are allowed up to 2 absences from recitation, no questions asked, without any impact on your grade.

If you can't make your normal recitation time in a given week, you can opt to attend an alternative recitation (if you can make another recitation time). Just confirm with your TAs that you have been marked as attending recitation.

For more details about recitation, please see the recitation schedule.

4. Religious Observations

I try my best to coordinate the course schedule around major holidays not built into the academic calendar (e.g., Rosh Hashanah, Yom Kippur, Purim, Diwali, Holi, Eid al-Fitr). However, sometimes it's not always possible.

In accordance with NYU's policy on religious observance, iIf you wish to take part in religious observances that occur during this academic term and conflict with participation in this course, you should reach out to me within the first couple weeks of class to discuss accommodations.

For example, I know Exam 1 overlaps with Ramadan this year. My assumption is that fasting students will find it beneficial to take the exam earlier in the day (closer to Suhoor). If that's the case, students observing Ramadan can reach out to me about taking the exam earlier the same day through the Moses Center.

5. Email policy

I will answer emails as soon as possible between the hours of 9 am and 5pm Monday through Friday. Messages outside of these hours will be answered as soon as possible, typically within 24 hours (or not till Monday if over the weekend).

In your email subject line, please include "PSYCH 11" before your subject line. E.g., "Psych 11 - Question about Exam 1". I teach multiple classes (and otherwise get lots of emails), so this subject line will make it easier to identify!

Before sending an email, please check if your question is answered on the syllabus (e.g., in the FAQ section). If your question pertains to the content of the course (e.g., you're confused about something from the book or said in lecture), these questions are usually better answered in-person (with either me or your TA) unless it's a simple yes/no question.

6. Academic Integrity/Plagiarism

Academic Integrity, Plagiarism, and Cheating (adapted from the website of the College of Arts and Science, https://cas.nyu.edu/content/nyu-as/cas/academic-integrity.html):

Academic integrity means that the work you submit is original. Obviously, bringing answers into an examination or copying all or part of a paper straight from a book, the Internet, or a fellow student is a violation of this principle. But there are other forms of cheating or plagiarizing which are just as serious — for example, presenting an oral report drawn without attribution from other sources (oral or written); writing a sentence or paragraph which, despite being in different words, expresses someone else's idea(s) without a reference to the source of the idea(s); or submitting essentially the same paper in two different courses (unless both instructors have given their permission in advance). Receiving or giving help on a take-home paper, examination, or quiz is also cheating, unless expressly permitted by the instructor (as in collaborative projects).

Most importantly, cheating only really serves to cheat yourself out of a learning opportunity. There are many opportunities to make up for a less-than-ideal grade, so don't let the possibility of a bad grade tempt you to cheat.

We will do our best to make clear the line between academic dishonesty and collaborative learning. But if you have any questions about where this line is, it is your responsibility to ask me or your TA.

7. Use of AI (e.g., ChatGPT)

In general, I highly discourage students from using AI as current research (e.g., this MIT study) suggests that using AI may harm learning. One of the greatest and most versatile skills you can develop during college is the ability to learn, and you cheat yourself of this opportunity when over-relying on AI.

Using AI tools to generate work is considered plagiarism and will result in disciplinary action. For example, submitting a write-up of data written by AI is not allowed.

While discouraged, students may opt to use AI to help in initial idea generation, creating outlines, or writing R code. This is allowed but any AI-generated content must be properly attributed. If you use AI in the course of completing an assignment, please include a statement at the end of your paper/assignment describing how you incorporated AI in your writing or coding process.

Additionally, you are responsible for understanding what the code or ideas generated mean.

8. Disability Disclosure Statement

I encourage students with disabilities, including non-visible disabilities like chronic diseases or learning disabilities, to reach out to me early in the semester to discuss accommodations.

You should also contact the Moses Center as soon as possible to verify eligibility for academic accommodations. Accommodation statements should be shared with me as early as possible, but if there is a change to the course structure that would better serve you (and your fellow students), please let me know!

Henry and Lucy Moses Center for Student Accessibility
Telephone: 212-998-4980
Website: https://www.nyu.edu/students/communities-and-groups/student-accessibility.html
Email: mosescsa@nyu.edu

9. Wellness Statement

In a large, complex community like New York University, it's vital to reach out to others, particularly those who are isolated or engaged in self-destructive activities. Student wellness (https://cas.nyu.edu/content/nyu-as/cas/academic-programs/student-wellness.html) is the responsibility of us all.

The NYU Wellness Exchange is the constellation of NYU's programs and services designed to address the overall health and mental health needs of its students. Students can access this service 24 hours a day, seven days a week: wellness.exchange@nyu.edu; (212) 443-9999. Students may also call the Wellness Exchange hotline (212-443-9999) or the NYU Counseling Service (212-998-4780) to make an appointment for single session, short-term, or group counseling sessions.

10. Respect for diversity

It is my intent that students from diverse backgrounds and perspectives be well served by the course and that the diversity—in its many forms—that students bring to this class be viewed as a strength. It is my intent to present only materials and activities that are respectful to diversity.

Additionally, as will be noted in this course, the field of statistics has a difficult past to contend with. Historically, the field has been dominated by certain perspectives, with many of the "great" statisticians developing statistical tests and techniques with the intent to provide quantitative support for ideas like eugenics and racial/class prejudice.

These statistical tests are still used today and are many of the tests that will be covered in this course. Are these tests inherently bad? No. But it's important to consider what data we gain when we focus on questions around differences in group means and what information we lose. I hope we can acknowledge these limitations, while still thoughtfully engaging with the foundational material of the field.

What you will do in this course

1) Exams (30%)

There will be two exams in this course. Exams will consist primarily of multiple-choice questions. Exams will be in-class and timed (75 minutes). You can use your notes (but not the textbook). Notes for the exam must be synthesized onto a single 8.5 x 11" sheet of paper.

An optional final exam at the end of semester can be taken to replace one of your lowest exam scores. The final exam is cumulative.

2) Homework (40%)

Assignments will focus on applying the statistical concepts and R programming skills learned, including data analysis, interpretation, and presentation of results. Assignments are designed to build cumulative skills and prepare students for more comprehensive projects. The lowest two assignment scores will be dropped.

3) Data Reports (30%)

Data reports build off homework assignments by asking students to complete analyses in a less guided nature than in the homeworks. For these data reports, students will be asked to analyze a provided dataset using appropriate statistical methods implemented in R. Assessment will be based on selecting and executing relevant statistical analyses, interpreting the results, and presenting their findings in a written report.

Course schedule

The below schedule reflects our tentative course schedule. This is roughly broken down by theme. Depending on the pace of our class, I reserve the right to make changes to the course schedule. Updates will be made to the syllabus if changes are made and reflected on Brightspace.

Week	Date	Topic	Readings	Assignment due
1	Jan 21	1. Introduction to course		
2	Jan 26	2. Introduction to R	RDS Chap 2 + 7	
	Jan 28	3. Data and measurement	E Chap 8 + LSR Chap 2	
3	Feb 2	4. Describing data	LSR Chap 5	**Homework 1 due Feb 6**
	Feb 4	5. Visualizing data	LSR Chap 6 + RDS Chap 9	
4	Feb 9	6. Data wrangling in R I	RDS Chap 3 + 5	**Homework 2 due Feb 13**
	Feb 11	7. Data wrangling in R II		

5	Feb 17	8. Probability	LSR Chap 9	**Homework 3 due Feb 20**
	Feb 18	9. Sampling	LSR Chap 10	
6	Feb 23	10. Estimation	LSR Chap 10	**Homework 4 due Feb 27**
	Feb 25	11. Hypothesis testing	LSR Chap 11	
7	Mar 2	12. Hypothesis testing: *t*-test	LSR Chap 13	**Homework 5 due Mar 6**
	Mar 4	13. Statistical power and effect size		
8	Mar 9	14. Hypothesis testing: Chi-square *(won't be on Exam 1)*	LSR Chap 12	
	Mar 11	**Exam 1**		
9	Mar 16	**No Class: Spring Break**		
	Mar 18			
10	Mar 23	15. Hypothesis testing: One-way ANOVA	LSR Chap 14	**Homework 6 due Mar 27**
	Mar 25	16. Hypothesis testing: Factorial ANOVA	LSR Chap 15	
11	Mar 30	17. Hypothesis testing: Non-parametric tests		**Homework 7 due Apr 2**
	Apr 1	18. Statistical models: Correlation and linear regression	LSR Chap 15	
12	Apr 6	19. Statistical models: Multiple regression I	LSR Chap 15	**Data Report 1 due Apr 10**
	Apr 8	20. Statistical models: Multiple regression II		
13	Apr 13	21. Statistical models: Mixed-effects		**Homework 8 due Apr 17**
	Apr 15	22. Statistical models: Logistic regression		
14	Apr 20	23. Statistical models: Path Models		**Homework 9 due Apr 23**
	Apr 22	24. Multivariate Statistics I		
15	Apr 27	25. Multivariate Statistics II		
	Apr 29	26. Meta-analyses	E Chap 16	
16	May 4	**Exam 2**		**Data Report 2 due May 8** (hard deadline is May 13 – a little extra time for those opting to take the final exam)
	May 11	**Final Exam** (Optional) 10 am - 11:50		

When are assignments due? Do you accept late assignments?

Assignments are due at 11:59 pm on the date listed above.

In this class, I adopt soft(ish) deadlines: there is a built-in grace period for all due dates (unless otherwise noted above). This means that for all of the assignments (which are due Friday night at 11:59 pm), you have until Monday morning at 7 am to turn them in without penalty.

Please aim for the deadline listed on the syllabus, but know there is some grace in the final submission in case life happens and something comes up! I will not grant an extension beyond the grace period given.

Frequently asked questions

- Related to grades
  - What is the grading scale for this course? Do you round grades?
  - My grade is an 89.4? Will you round it up an A-?
  - Is there extra credit in this course?
- Related to exams
  - Will there be a study guide for exams?
  - Will there be practice questions for the exams?
  - How should I study for the exam?
  - Can I use notes on the exam?
  - Can you print out my notes for me?
  - Do I need a Scantron? A pencil?
  - Are the exams cumulative?
  - Are the exams curved?
  - Do I need to take the final exam?
  - Do you offer make-up exams?
  - How do I see my exam after it's been graded?
  - I did poorly on an exam. Can I retake it?
- Related to lecture
  - Will you post lecture slides?
  - Do I need to email you if I am unable to attend lecture?
  - Will lectures be recorded?
  - Where are the recordings available?
  - I can't access the recordings. How do I gain access?
  - Can you download the recording and send it to me?
- Related to recitation
  - Can I attend a different recitation than my official recitation time?
  - Do I need to email my TA that I am missing recitation?
  - Do I need to email you that I am missing recitation?
  - Can I do something else to make up for missed recitations?
- Related to assignments
  - When are assignments due?

Are late assignments accepted?

Related to grades

What is the grading scale for this course? Do you round grades?
Your numerical grade will be rounded to the nearest integer and turned into a letter grade as follows:

93-100 A	90-92 A-	87-89 B+	83-86 B	80-82 B-	
77-79 C+	73-76 C	70-72 C-	67-69 D+	60-66 D	0-59 F

For example, an 89.5 will be rounded up to a 90; an 89.4 will not. This rounding will be done automatically. You do not need to request it.

My grade is an 89.4? Will you round it up an A-?
No. The cutoff for an A- is a 90. Scores of 89.5 will be rounded up.

Is there extra credit in this course?
Generally no, but there are numerous opportunities to make up lost points (e.g., lowest exam score dropped, lowest research critique dropped). Occasionally, I may offer an extra credit point or two on exams.

If I have not offered extra credit publicly to the whole course (e.g., on the syllabus, through a course announcement on Brightspace), it doesn't exist.

Related to exams

Will there be a study guide for exams?
Yes, I will provide a document with topics you should be familiar with. But it's up to you to create those notes on each topic! Making your own notes is a part of the encoding process, meaning you will learn the material better if you are able to put the content into your own words!

The study guide is pretty exhaustive. That is, if you knew everything on the study guide, you would be able to get an A on the exam.

Will there be practice questions for the exams?
There will be review during your recitations.

How should I study for the exam?
Come to lecture, read the textbook, go over your notes, work through the study guide, meet with either me or one of the TAs to go over questions you have, create a study group with your fellow students.

Can I use notes on the exam?
Yes, I allow one page of notes, 8.5 x 11 inches. Front and back. You can type your notes, handwrite them, whatever. No specific font requirements. I recommend consolidating your notes into your own words.
Your notes must be printed out (e.g., no notes on phones, laptops, etc.)

Can I use my phone/tablet/computer to access my notes during the exam?
No, your notes must be printed out. You can print your notes out using the university printing services available here:
https://www.nyu.edu/life/information-technology/computing-support/desktop-mobile-and-printing/nyu-print-service.html

There's also a Staples near campus.

Can you print out my notes for me?
No, but you can print your notes out using the university printing services available here:
https://www.nyu.edu/life/information-technology/computing-support/desktop-mobile-and-printing/nyu-print-service.html

Do I need a Scantron? A pencil?
No Scranton needed. You can use whatever writing implement you like best (pen, pencil, whatever).

Are the exams cumulative?
No, except for the final. However, a lot of topics in this course build off each other.

Are the exams curved?
In the sense that only X% of students can get an A, X% can get a B, and so on, no. Theoretically, everyone in the class can get an A on the exam.

In the other sense (i.e., adjusting the total points an exam is out of after the fact), I don't write exams with the intention of them being curved, but if it appears the exam was harder than intended, then I might curve the exam by a point or two.

Do I need to take the final exam?
If you are happy with your score on the first two exams, you do not need to take the final (as your lowest exam score is dropped, so the "0" for the final would be dropped).

Do you offer make-up exams?
We will offer two make-up exam dates. These make-up exams will happen the week after the original exam during lecture time.

These make-up exams are intended for students experiencing illnesses, those who have a conflict with a religious holiday, or those otherwise dealing with an emergency. Please do not abuse this policy if one of the following does apply to you. The rooms we have available to hold make-up exams are small and cannot accommodate a large number of students.

Please reach out to me directly for a make-up exam request (please cc theTA on the email). Your request must come before the exam (when reasonably possible).

If you do not come to your scheduled make-up exam, you can make up for the missed exam by taking the final exam.

How do I see my exam after it's been graded?
Your TA will go over exams during recitation. If you cannot attend recitation that week, please contact your TA to set up a time to look over your exam or visit them during their scheduled office hour.

I did poorly on an exam. Can I retake it?
No, but good news! Your lowest exam grade is dropped. There are three total exams and only two of them count toward your final grade.

Related to lecture

Will you post lecture slides?
Yes! Lecture slides will be posted on Brightspace before (or at the start of) class. I cannot guarantee they will be posted in advance of the start of class.

Do I need to email you if I am unable to attend lecture?
No. However, if you anticipate missing multiple classes due to extenuating circumstances (e.g., an illness) that will significantly impact your ability to do well in class, please feel free to reach out to discuss your situation to ensure you stay on track.

Will lectures be recorded?
Yes.

Where are the recordings available?
You can find them on Brightspace by going to the "Zoom" tab and clicking on "Cloud recordings".

I can't access the recordings. How do I gain access?
Make sure you are logged into both Brightspace and your NYU Zoom account.

You must log into NYU Zoom before being able to access the videos on Brightspace. To log into your NYU Zoom account, please visit https://nyu.zoom.us/.

Lecture recordings are only accessible when logged into your NYU Zoom account. After you log into your NYU Zoom account, you may need to refresh the Brightspace page for the videos to load (if Brightspace was already open before logging into NYU Zoom).

If you are still unable to access the recordings, please contact IT to resolve your issue.

If you reach out to me directly, I will point you to these instructions on the syllabus–and if these instructions don't work, I will direct you to IT. Sorry, there's nothing else I can do on my end to fix your issue! There's no special permission that I need to check off to make the videos available to you.

Consider closing and reopening the browser or refreshing the page if the issues persist.

Can you download the recording and send it to me?
No. I don't feel comfortable with that.

Related to recitation

Can I attend a different recitation than my official recitation time?
Yes, that should generally be fine.

Do I need to email my TA that I am missing recitation?
No, unless you are dealing with a chronic health condition or extenuating circumstance that may impact your ability to attend recitations regularly.

Do I need to email you that I am missing recitation?
No. Please direct your questions about recitations to your TA!

Related to assignments

When are assignments due?
At 11:59 pm at the date listed on the syllabus unless otherwise stated. The dates are also listed on Brightspace.

If there is a discrepancy between the syllabus and Brightspace, please use the date listed on Brightspace.

Are late assignments accepted?
Yes! But with a penalty.

Assignments submitted after the deadline will incur a 10% penalty and assignments submitted 72 hours after the deadline will incur a 25% penalty.

For the soft deadlines, this means that for all of the assignments that are due on a Friday night at 11:59 pm, you have until Monday morning at 7 am to turn them in without penalty. After that point, the assignment is late and will incur the 10% penalty. If you haven't turned it in by Thursday at 7 am, the penalty increases to 25%.

If you anticipate turning in an assignment late for a reason you feel that you should not be penalized for, please email me and your TA before the due date to discuss your situation.

Late assignments may not be graded as quickly as assignments submitted on time.

Wait, so I really can submit an assignment from earlier in the semester?
Yes! Turning in work late is better than submitting nothing. There is a 10 - 25% penalty depending on how late the assignment is.